

ZAVITINSK, Russian FederationWMO: **315270**Lat: **50.117N**Lon: **129.467E**Elev: **786**StdP: **14.28**Time Zone: **9.00 (E09)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.6	-25.2	-33.9	0.8	-27.3	-30.8	1.0	-24.3	14.5	-4.5	12.5	-1.6	1.4	250	0.537

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.7	84.8	69.6	82.0	68.4	79.4	67.1	72.9	80.5	71.0	78.6	69.4	76.8	4.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.6	116.1	76.7	68.5	107.7	75.0	66.7	101.1	73.6	37.2	80.4	35.5	78.4	34.1	77.2	91.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
15.9	13.4	11.4	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-33.4	90.1	4.6	2.8	-36.7	92.2	-39.5	93.8	-42.0	95.4	-45.4	97.4		
			-33.5	76.8	4.5	4.5	-36.7	80.1	-39.3	82.7	-41.8	85.2	-45.1	88.5		
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Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	32.9	-9.3	-0.6	16.7	38.5	53.5	64.6	69.8	65.8	53.6	36.0	11.3	-7.7	
	DBStd	29.54	7.70	8.80	10.98	8.79	7.65	6.02	4.28	5.40	7.40	9.17	12.09	8.74	
	HDD50	8135	1839	1416	1033	361	49	0	0	1	46	441	1161	1788	
	HDD65	12050	2304	1836	1498	797	363	79	10	54	347	898	1611	2253	
	CDD50	1878	0	0	0	14	159	437	615	490	155	8	0	0	
	CDD65	318	0	0	0	0	8	66	160	79	5	0	0	0	
	CDH74	2341	0	0	0	10	135	610	1045	500	40	1	0	0	
	CDH80	484	0	0	0	1	23	161	224	73	2	0	0	0	
Wind	WSAvg	4.2	3.1	3.6	4.8	5.6	5.3	3.8	3.5	3.5	4.2	5.0	4.3	3.2	
Precipitation	PrecAvg	23.4	0.3	0.3	0.6	1.2	2.3	3.5	4.9	4.8	3.0	1.5	0.8	0.6	
	PrecMax	35.0	0.8	1.1	1.5	3.7	4.8	8.4	9.0	11.6	4.9	3.8	1.8	1.7	
	PrecMin	18.7	0.0	0.0	0.0	0.0	0.8	0.5	1.2	1.7	0.7	0.2	0.2	0.2	
	PrecStd	4.1	0.2	0.3	0.5	0.9	1.2	1.6	2.1	2.0	1.3	0.9	0.4	0.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.2	29.5	47.6	74.4	83.2	89.7	88.3	85.0	78.5	67.5	44.9	21.8	
		MCWB	13.7	24.7	37.0	51.7	59.1	70.0	71.9	71.3	65.7	51.6	37.7	19.4	
	2%	DB	9.7	24.0	41.6	63.8	77.9	84.4	85.3	82.0	74.2	60.3	38.2	13.8	
		MCWB	8.2	20.8	33.1	47.0	57.3	67.4	71.5	70.0	62.1	48.4	33.0	12.4	
	5%	DB	5.8	18.6	37.3	58.3	73.0	80.5	82.4	79.2	70.7	54.8	33.3	8.9	
		MCWB	4.5	16.0	30.9	44.3	55.0	65.6	70.2	68.4	59.8	44.1	29.7	7.5	
	10%	DB	2.6	13.6	33.5	53.5	68.4	77.0	79.7	76.5	66.8	50.3	29.4	5.4	
		MCWB	1.5	11.5	28.5	41.2	53.0	64.7	69.1	67.0	57.4	41.9	26.5	4.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.1	25.2	38.0	53.0	62.5	73.4	78.1	74.9	67.6	53.4	38.8	19.8
			MCDB	14.9	28.8	46.7	71.8	75.8	86.0	82.4	81.0	74.9	63.5	43.7	21.9
2%		WB	8.2	20.7	34.0	48.0	59.4	69.6	73.5	72.3	64.1	49.7	33.1	12.4	
		MCDB	9.6	23.9	39.9	61.2	72.2	80.6	80.8	78.5	71.9	58.3	36.8	13.7	
5%		WB	4.6	15.8	31.3	45.0	57.2	67.5	71.7	70.2	61.8	45.7	30.1	7.7	
		MCDB	5.7	18.4	36.7	56.6	68.8	77.2	79.3	76.5	68.4	53.0	33.4	8.9	
10%		WB	1.6	11.5	28.6	42.2	55.0	65.7	70.2	68.4	59.2	42.5	26.5	4.3	
		MCDB	2.7	13.5	33.5	52.7	66.5	75.0	78.0	74.7	64.3	49.7	29.3	5.4	
Mean Daily Temperature Range	5% DB	MDBR	16.7	19.3	19.0	18.9	20.4	18.5	16.7	17.3	19.5	17.9	16.3	16.0	
		MCDBR	17.9	21.3	19.4	26.0	27.9	24.7	21.0	21.0	24.7	25.2	18.0	16.6	
		MCWBR	16.7	18.7	13.5	14.8	12.8	11.3	10.3	10.6	13.3	15.0	13.8	15.5	
	5% WB	MCDDBR	17.9	21.2	18.1	24.1	23.1	21.0	17.4	18.0	20.7	21.9	16.7	16.5	
		MCWBR	16.8	18.8	12.8	14.4	12.6	11.2	10.1	10.9	12.8	14.4	13.0	15.5	
Clear-Sky Solar Irradiance	taub	0.273	0.303	0.382	0.536	0.588	0.499	0.480	0.431	0.402	0.416	0.338	0.279		
	taud	2.134	2.054	1.888	1.705	1.684	1.995	2.104	2.233	2.221	2.005	2.074	2.091		
	Ebn at Noon	237	258	252	224	220	244	247	254	248	209	202	211		
	Edh at Noon	25	35	50	69	73	54	48	40	36	37	27	23		
All-Sky Solar Radiation	RadAvg	544	936	1418	1576	1722	1865	1741	1530	1220	797	520	422		
	RadStd	13	19	86	170	137	227	128	102	111	44	31	23		